

1. ① $\bar{A} \cup \bar{B} \cup \bar{C}$; ② $\bar{A}\bar{B} \cup \bar{A}\bar{C} \cup \bar{B}\bar{C}$; ③ $A \cup B \cup C$;
 ④ $AB \cup AC \cup BC$; ⑤ = ②; ⑥ $A\bar{B}\bar{C} \cup \bar{A}B\bar{C} \cup \bar{A}\bar{B}C$
 ⑦ $AB\bar{C} \cup A\bar{B}C \cup \bar{A}B\bar{C}$; ⑧ $AB\bar{C} \cup \bar{A}B\bar{C}$; ⑨ $\bar{A}\bar{B}\bar{C}$; ⑩ $\bar{A}\bar{B}\bar{C}$.

2. $\bar{A}\bar{B}$: 至 A, B 至少有一个不发生; $A \cap B = A, C$ 同时发生或 B 发生.

① $P(\bar{A}\bar{B}) = P(A - B) = P(A) - P(AB) = 1 - P(\bar{A}) = 1 - a$.

② $0.7 = P(A \cup B) = P(A) + P(B) - P(AB) = P(A) + P(B) - P(A)P(B)$
 $= 0.3 + P(B) - 0.3P(B), \therefore P(B) = \frac{4}{7}, P(AB) = P(A)P(B) = \frac{12}{7}$.

③ $P(AB) = P(B)P(A|B) = 0.56, P(A \cup B) = P(A) + P(B) - P(AB)$
 $= 0.84$.
 $P(B|A) = \frac{P(AB)}{P(A)} = \frac{14}{15}$.

④ $P(\bar{A}\bar{B}) = P(\bar{A})P(\bar{B}) = 0.49$; $P(B|A) = P(B) = 0.7$.

⑤ $0.3 = P(A - B) = P(A) - P(AB) \Rightarrow P(AB) = 0.4, P(B|\bar{A}) = \frac{P(B\bar{A})}{P(\bar{A})}$
 $= \frac{P(B - A)}{1 - P(A)} = \frac{P(B) - P(AB)}{1 - P(A)} = \frac{0.5 - 0.4}{0.3} = \frac{1}{3}$

⑥ $P(B - A) = P(B) - P(A) = 0.3$.

⑦ $P(\bar{A}\bar{B}\bar{C}) = 1 - P(A \cup B \cup C) = 1 - [P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC)]$
 $= 1 - (\frac{1}{4} + \frac{1}{4} + \frac{1}{4} - 0 - \frac{1}{8} - \frac{1}{8} + 0) = \frac{7}{12}$

⑧ ① $1 - (1 - p)^3 = \frac{37}{64} \Rightarrow (1 - p)^3 = \frac{27}{64} \Rightarrow 1 - p = \frac{3}{4} \Rightarrow p = \frac{1}{4}$

② 女: 0, 男: 1. $\begin{cases} \text{有 1 个女孩: } 011, 101, 110; \\ \text{有 2 个女孩: } 100, 010, 001 \\ \text{有 3 个女孩: } 000 \end{cases}$

总数: 7 种, 含有男孩的有 6 种, $p = \frac{6}{7}$.



4. ① $1 = \int_0^1 c x^2 dx = \frac{c}{3} \Rightarrow c = 3$; ②
$$\begin{cases} F(0) = 0 \\ F(0+0) = 1+b \Rightarrow b = -1 \\ F(0) = F(0+0) \end{cases}$$

5. ① $1 = \frac{1}{2a} + \frac{3}{4a} + \frac{5}{8a} = \frac{15}{8a} \Rightarrow a = \frac{15}{8}$

② $1 = (-2c^2 + c + 2c^2) = 2c \Rightarrow c = \frac{1}{2}$

6. X 的分布律:

X	-2	-1	0	1
P	0.2	0.15	0.25	0.4

$$E(X) = -2 \times 0.2 + (-1) \times 0.15 + 0 \times 0.25 + 1 \times 0.4 = -0.15$$

Y 的分布律:

Y	0	1	2
P	0.15	0.45	0.4

, $E(Y) = 0 \times 0.15 + 1 \times 0.45 + 2 \times 0.4 = 1.25$.

7. $F(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$; $Y \sim N(a\mu+b, a^2\sigma^2)$

8. $X \sim N(1, 4)$, $F(5) = P\{X \leq 5\} = P\left\{\frac{X-1}{2} \leq \frac{5-1}{2}\right\} = \Phi(2)$

$$P\{0 \leq X \leq 1.6\} = P\left\{\frac{0-1}{2} \leq \frac{X-1}{2} \leq \frac{1.6-1}{2}\right\} = \Phi(0.3) - \Phi(-0.5) \\ = \Phi(0.3) + \Phi(0.5) - 1$$

$$P\{|X-1| \leq 2\} = P\{-1 \leq X \leq 3\} = P\left\{-\frac{1-1}{2} \leq \frac{X-1}{2} \leq \frac{3-1}{2}\right\} \\ = \Phi(1) - \Phi(-1) = 2\Phi(1) - 1$$

(注: 试卷上会给出 $\Phi(x)$ 的具体值)

9. $p_1 < p_2$

10. $P\{X > 2\} = \frac{2}{3}$, $P\{X > 2\} = P\{Y = 3\} = \left(\frac{2}{3}\right)^3 = \frac{8}{27}$

11. a



12. $5e^{-4}$; 4; 4.

13. $P\{X \geq 3\} = 1 - P\{X < 3\} = 1 - e^{-8}(1+8+32)$
 $= 1 - 41e^{-8}$

14. 设 Y 为 200 小时内损坏的电子元件数,

$P\{200 \text{ 小时内一只电子元件损坏}\} = P\{X \leq 200\} = F(200) = 1 - e^{-\frac{1}{3}}$

$P\{200 \text{ 小时内三只电子元件中至少有一只损坏}\} = 1 - [P\{e^{-\frac{1}{3}}\}]^3 = 1 - e^{-1}$

(注: $Y \sim B(3, 1 - e^{-\frac{1}{3}})$.)

15. $1 - \frac{C_5^4 \cdot C_5^1 \cdot C_5^1 \cdot C_5^1 \cdot C_5^1}{C_{10}^5} = 1 - \frac{80}{210} = \frac{13}{21}$

16. 泊松分布; 指数分布; χ^2 分布.

17. $f(x, y) = \begin{cases} \pi, & x^2 + y^2 \leq 1 \\ 0, & \text{其他} \end{cases}$

18. $E(Z) = E(3X-2) = 3E(X) - 2 = 3 \times 2 - 2 = 4$

$D(Z) = D(3X-2) = 9D(X) = 18$

$\text{COV}(X, Z) = \text{COV}(X, 3X-2) = 3\text{COV}(X, X) = 3D(X) = 6$

$\rho_{XZ} = \frac{\text{COV}(X, Z)}{\sqrt{D(X) \cdot D(Z)}} = \frac{6}{\sqrt{2 \times 18}} = 1.$

19. $E(3X^2 - 2X + 4) = 3EX^2 - 2EX + 4 = 3[DX + (EX)^2] - 2EX + 4$
 $= 37$

20. $\frac{1}{10}$; 44.



21. 1; 1.

22. $X+Y=n$, $\therefore \rho_{XY} = -1$

23. $3 = D(X+Y) = DX + DY + 2\text{cov}(X, Y) = 0.96 + 4 + 2\text{cov}(X, Y)$

$\therefore \text{cov}(X, Y) = -0.98$. $\therefore \rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{DX \cdot DY}} = \frac{-0.98}{4\sqrt{0.24}} = -0.25$

24. $\bar{X} \sim N(\mu, \sigma^2/n)$. 25. $\frac{1}{9}$.

26. $P\{|\bar{X}-3| \geq 2\} \leq \frac{\sigma^2}{204^2}$; 27. $\frac{1}{2}$ 28. $\Phi(x)$, 29. $\Phi(x)$.

30. 6. 31. 0 32. 1. 33. $\frac{1}{20}$; 2.

34. $t(n)$; 35. $t(4)$. 36. $\frac{1}{2}$, 37. 0.45.



$$63. 1 - 0.8 \times 0.5 \times 0.4 = 0.91$$

$$64. 1 - 0.75 \times 0.7 \times 0.6 = 0.685$$

$$65. (1) p = \frac{3}{6} = \frac{1}{2}, (2) p = \frac{4 \times 4 \times 4}{6^3} = \frac{5}{14}$$

$$66. 1 - (1-p)^3 = 8/9, \therefore p = \frac{1}{3} \quad 1 - p^3 = \frac{1}{4} \quad 1-p = \frac{1}{4}, p = 1 - \frac{1}{4}$$

$$67. (1) \frac{C_6^3 \cdot C_{36}^{48} - 364^{48}}{365^{48}}, (2) 1 - \frac{334^{50}}{365^{50}}$$

$$68. p(A_1 | \bar{A}_3) = \frac{p(A_1 \bar{A}_3)}{p(\bar{A}_3)} = \frac{p(A_1)}{1 - p(A_3)} = \frac{0.6}{0.9} = \frac{2}{3}$$

$$69. (1) p\{x < 4\} = p\{x = 3\} = \frac{1}{C_6^3} = \frac{1}{20}$$

$$(2) p\{x = 3\} = \frac{1}{C_6^3} = \frac{1}{20}, p\{x = 4\} = \frac{C_3^2}{C_6^3} = \frac{3}{20}$$

$$p\{x = 5\} = \frac{C_3^2}{C_6^3} = \frac{6}{20} = \frac{3}{10} \quad p\{x = 6\} = \frac{C_3^2}{C_6^3} = \frac{1}{2}$$

也可:

x	3	4	5	6
p	1/20	3/20	3/10	1/2

70. 设事件 $A = \{\text{甲发球失误}\}$, $B_i = \{\text{乙第 } i \text{ 次发球失误}\}$, $i=1, 2$.

$$\text{则 } p(B_1) = 0.3, p(A | \bar{B}_1) = 0.4, p(B_2 | \bar{A} \bar{B}_1) = 0.5$$

$$p\{\text{乙输1分}\} = p(B_1 \cup \bar{B}_1 \bar{A} B_2) = p(B_1) + p(\bar{B}_1 \bar{A} B_2)$$

$$= p(B_1) + p(\bar{B}_1) p(\bar{A} | \bar{B}_1) p(B_2 | \bar{B}_1 \bar{A}) = 0.3 + 0.7 \times 0.6 \times 0.5 = 0.51$$

71. 设 $A_1 = \{\text{甲射中目标}\}$, $A_2 = \{\text{乙射中目标}\}$, $B_1 = \{\text{丙射中目标}\}$

$B_2 = \{\text{丁射中目标}\}$, $W = \{\text{完成使命}\}$

(1) 甲丙, 乙丁搭配

$$p(W) = p\{\text{甲丙中} \cup \text{乙丁中}\} = p\{\text{甲丙中}\} + p\{\text{乙丁中}\} - p\{\text{都中}\}$$

$$= 0.9 \times 0.7 + 0.8 \times 0.6 - 0.9 \times 0.7 \times 0.8 \times 0.6 = 0.8706$$

$$(2) \text{甲丁, 乙丙搭配}, p(W) = 0.8 \times 0.6 + 0.8 \times 0.7 - 0.8 \times 0.8 \times 0.7 \times 0.6 = 0.7976, \text{甲丙, 乙丁搭配更好.}$$



12. 设事件 A_1, A_2 分别为甲、乙系统有效, 则 $P(A_1) = 0.92$,
 $P(A_2) = 0.93$, $P(A_2|\bar{A}_1) = 0.85$.

$$(1) P\{\text{至少有一个有效}\} = P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 A_2)$$

$$\because P(A_2|\bar{A}_1) = \frac{P(A_2) - P(A_1 A_2)}{1 - P(A_1)} = 0.85, \therefore P(A_1 A_2) = 0.862$$

$$\therefore P(A_1 \cup A_2) = 0.988$$

$$(2) P(A_1|\bar{A}_2) = \frac{P(A_1) - P(A_1 A_2)}{1 - P(A_2)} = \frac{0.92 - 0.862}{0.07} = 0.82857$$

13. $X \sim B(3, \frac{1}{5})$, $P\{X=k\} = C_3^k (0.2)^k 0.8^{3-k}$, $k=0, 1, 2, 3$.

$$\text{或 } P\{X=0\} = 0.8^3 = 0.512, P\{X=1\} = C_3^1 \times 0.2 \times 0.8^2 = 0.384$$

$$P\{X=2\} = C_3^2 \times 0.2^2 \times 0.8 = 0.096, P\{X=3\} = 0.008$$

或

X	0	1	2	3
P	0.512	0.384	0.096	0.008

此为分布律的三种形式.

14. 设事件 A_i 分别为第 i 条流水线, $i=1, 2, 3$. $B = \{\text{取到不合格品}\}$

$$P(A_1) = 20\%, P(A_2) = 50\%, P(A_3) = 30\%, P(B|A_1) = 0.05,$$

$$P(B|A_2) = 0.04, P(B|A_3) = 0.03$$

$$(1) P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)$$

$$= 0.2 \times 0.05 + 0.5 \times 0.04 + 0.3 \times 0.03 = 0.039$$

$$(2) P(A_1|B) = \frac{P(A_1)P(B|A_1)}{P(B)} = \frac{0.01}{0.039} = 0.2564.$$

$$15. (1) P\{\text{取到次品}\} = \frac{1}{2} \times 2\% + \frac{1}{4} \times 2\% + \frac{1}{4} \times 4\% = 0.025$$

$$(2) P\{\text{第1条}|\text{次品}\} = \frac{0.01}{0.025} = 0.4.$$



6. 设事件 A_i 分别为含有 0, 1, 2 件次品, $i=0, 1, 2$.

事件 $B = \{\text{产品通过检验}\}$, $P(A_0) = P(A_1) = P(A_2) = \frac{1}{3}$,

$P(B/A_0) = 1$, $P(B/A_1) = \frac{9}{10}$, $P(B/A_2) = \frac{4}{5}$

$$(1) P(B) = P(A_0)P(B/A_0) + P(A_1)P(B/A_1) + P(A_2)P(B/A_2) \\ = \frac{1}{3} \times (1 + 0.9 + 0.8) = 0.9.$$

$$(2) P(A_1/B) = \frac{P(A_1\bar{B})}{1 - P(B)} = \frac{P(A_1) - P(A_1)P(B/A_1)}{1 - P(B)} = \frac{P(A_1) - P(A_1)P(B/A_1)}{1 - P(B)} \\ = \frac{1}{3}$$

7. 设事件 $A = \{\text{发报台发出信号} a\}$, $B = \{\text{收报台收到信号} a\}$

$P(A) = 0.6$, $P(\bar{A}) = 0.4$, $P(B/A) = 0.8$, $P(\bar{B}/A) = 0.2$

$P(B/\bar{A}) = 0.1$, $P(\bar{B}/\bar{A}) = 0.9$

$$(1) P(B) = P(A)P(B/A) + P(\bar{A})P(B/\bar{A})$$

$$= 0.6 \times 0.8 + 0.4 \times 0.1 = 0.52$$

$$(2) P(A/B) = \frac{P(A)P(B/A)}{P(B)} = \frac{0.48}{0.52} = 0.923.$$

78. $P\{x=1\} = P\{x=2\}$, $x \sim p(\lambda)$, $\therefore \frac{\lambda^k}{k!} e^{-\lambda} = P\{x=k\}$, $k=1, 2$

$$\therefore \frac{\lambda^1}{1!} e^{-\lambda} = \frac{\lambda^2}{2!} e^{-\lambda} \Rightarrow \lambda = 2, \therefore x \sim p(2).$$

设 $Y = \{\text{没有印刷错误的页数}\}$, $P\{x=0\} = e^{-2}$

$Y \sim B(4, e^{-2})$. $P\{\text{至少有3页无印刷错误}\} = P\{Y=3\} + P\{Y=4\}$

$$= C_4^3 (e^{-2})^3 (1 - e^{-2}) + C_4^4 (e^{-2})^4 = 4e^{-6} + e^{-8}$$



$$X \sim e(1000) = e(\frac{1}{1000}). \quad F(x) = \begin{cases} 1 - e^{-\frac{1}{1000}x}, & x > 0 \\ 0, & x \leq 0. \end{cases}$$

$$(1) P\{X > 1000\} = 1 - P\{X \leq 1000\} = 1 - F(1000) = e^{-1}$$

$$(2) P\{\text{至少有一足寿命大于1000小时}\} = 1 - (1 - e^{-1})^3$$

$$80. (1) 1 = \int_{-\infty}^{+\infty} f(x) dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} A \cos x dx = 2A, \therefore A = \frac{1}{2}$$

$$(2) P\{0 < X < \frac{\pi}{4}\} = \int_0^{\frac{\pi}{4}} \frac{1}{2} \cos x dx = \frac{1}{2} \sin x \Big|_0^{\frac{\pi}{4}} = \frac{\sqrt{2}}{4}$$

$$(3) f(x) = \begin{cases} 0, & x < -\frac{\pi}{2} \\ \frac{1+\sin x}{2}, & -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \\ 1, & x > \frac{\pi}{2} \end{cases}$$

$$81. (1) f(x) = \begin{cases} 0, & x < 1 \\ 2(x + \frac{1}{x} - 2), & 1 \leq x \leq 2 \\ 1, & x > 2. \end{cases}$$

$$(2) F(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{2}x^2, & 0 \leq x < 1 \\ 2x - \frac{1}{2}x^2 + 1, & 1 \leq x < 2 \\ 1, & x \geq 2. \end{cases}$$

$$82. (1) P\{X < 2\} = F(2) = \ln 2, \quad P\{0 < X \leq 3\} = F(3) - F(0) = 1$$

$$P\{2 < X < \frac{5}{2}\} = F(\frac{5}{2}) - F(2) = \ln \frac{5}{2} - \ln 2 = \ln \frac{5}{4}$$

$$(2) f_X(x) = F'_X(x) = \begin{cases} \frac{1}{x}, & 1 \leq x < e \\ 0, & \text{其他}. \end{cases}$$

$$83. f_Y(y) = \begin{cases} f_X[h^{-1}(y)] \cdot |h'(y)|, & \alpha < y < \beta \\ 0, & \text{其他} \end{cases}$$

$$= \begin{cases} \frac{y-8}{32}, & 8 < y < 16. \\ 0, & \text{其他}. \end{cases}$$



84. $f(x) = \begin{cases} \frac{1}{2}, & 0 < x < 2 \\ 0, & \text{其他.} \end{cases}$

$f_Y(y) = \begin{cases} \frac{1}{4}, & 1 < y < 5 \\ 0, & \text{其他.} \end{cases}$

85. $y = e^x, x = \ln y = h(y), h'(y) = \frac{1}{y}, y > 1.$

$f_X(x) = \begin{cases} \frac{1}{y^2}, & y > 1 \\ 0, & \text{其他.} \end{cases}$

86. $y = 2 \sin x - 1, \sin x = \frac{y+1}{2}, x = \arcsin \frac{y+1}{2} = h(y), -1 \leq y \leq 1.$

$h'(y) = \frac{\frac{1}{2}}{\sqrt{1 - \frac{(y+1)^2}{4}}} \cdot f_Y(y) = \begin{cases} \frac{\arcsin \frac{y+1}{2}}{2\pi^2 \sqrt{1 - \frac{(y+1)^2}{4}}}, & -1 \leq y \leq 1 \\ 0, & \text{其他} \end{cases}$

87.

X	-1	1
P	$\frac{1}{2}$	$\frac{1}{2}$

,

Y	1	2	3
P	$\frac{1}{6}$	$\frac{2}{12}$	$\frac{1}{4}$

(1) $EX = -1 \times \frac{1}{2} + 1 \times \frac{1}{2} = 0, EY = 1 \times \frac{1}{6} + 2 \times \frac{2}{12} + 3 \times \frac{1}{4} = \frac{25}{12}$

$EX^2 = 1 \times \frac{1}{2} + 1 \times \frac{1}{2} = 1, EY^2 = 1 \times \frac{1}{6} + 4 \times \frac{2}{12} + 9 \times \frac{1}{4} = \frac{57}{12}$

$DX = EX^2 - (EX)^2 = 1, DY = EY^2 - (EY)^2 = \frac{57}{12} - \frac{25^2}{12^2} = \frac{59}{144}$

(2) $EXY = -1 \times \frac{1}{6} - \frac{2}{3} + \frac{1}{2} + \frac{3}{4} = \frac{5}{12}, \text{cov}(X, Y) = EXY - EX \cdot EY = \frac{5}{12}.$

(3) $\rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{DX DY}} = \frac{\frac{5}{12}}{\sqrt{\frac{59}{144}}} = \frac{5}{\sqrt{59}}$



88. (1) X 的分布律:

X	0	1	2
P	0.3	0.35	0.35

, Y 的分布律:

Y	-1	0	2
P	0.4	0.3	0.3

$$(2) P\{X=0, Y=-1\} = 0.05 \neq P\{X=0\} \cdot P\{Y=-1\} = 0.12$$

$\therefore X$ 与 Y 不独立.

$$(3) P\{Y=-1 | X=0\} = \frac{P\{X=0, Y=-1\}}{P\{X=0\}} = \frac{0.05}{0.3} = \frac{1}{6}.$$

89. (X, Y) 的联合概率分布:

$X \backslash Y$	0	1	2
0	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{1}{9}$
1	$\frac{2}{9}$	$\frac{2}{9}$	0
2	$\frac{1}{9}$	0	0

X 的概率分布:

X	0	1	2
P	$\frac{4}{9}$	$\frac{4}{9}$	$\frac{1}{9}$

Y 的概率分布:

Y	0	1	2
P	$\frac{4}{9}$	$\frac{4}{9}$	$\frac{1}{9}$

$$90. (1) EX^2 = \int_{-\infty}^{+\infty} x^2 f(x) dx = \int_0^1 x^2 \cdot 3x^2 dx = \frac{3}{5}$$

$$(2) EX = \int_0^1 x \cdot 3x^2 = \frac{3}{4}, \quad DX = EX^2 - (EX)^2 = \frac{3}{5} - \frac{9}{16} = \frac{3}{80}$$

$$D(2X+100) = 4DX = \frac{3}{20}.$$



$$E(X) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x f(x, y) dx dy = \int_0^1 dx \int_0^x (1+y) dy = 4/5$$

$$E(Y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} y f(x, y) dx dy = \int_0^1 dx \int_0^x 12y^4 dy = 4/5$$

$$E(X^2) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x^2 f(x, y) dx dy = \int_0^1 dx \int_0^x x^2 (1+y) dy = 2/5$$

$$E(Y^2) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} y^2 f(x, y) dx dy = \int_0^1 dx \int_0^x y^2 (1+y) dy = 2/5$$

$$D(X) = E(X^2) - (E(X))^2 = \frac{2}{5} - \frac{16}{25} = \frac{2}{75}$$

$$D(Y) = E(Y^2) - (E(Y))^2 = \frac{2}{5} - \frac{16}{25} = \frac{2}{75}$$

$$E(XY) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy f(x, y) dx dy = \int_0^1 dx \int_0^x xy (1+y) dy = \frac{1}{5}$$

$$E(X^2Y^2) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x^2 y^2 f(x, y) dx dy = \int_0^1 dx \int_0^x x^2 y^2 (1+y) dy = \frac{2}{15}$$

$$D(XY) = E(XY)^2 - [E(XY)]^2 = \frac{2}{15} - \frac{1}{25} = \frac{1}{25}$$

$$\text{cov}(X, Y) = E(XY) - EX \cdot EY = \frac{1}{5} - \frac{4}{5} \times \frac{2}{5} = -\frac{1}{5}$$

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{D(X) \cdot D(Y)}} \quad (\text{注: 亦可先求边缘密度再求 } EX, EY \text{ 等})$$

92. (1) $E(X) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x f(x, y) dx dy = \int_0^1 dx \int_x^{2x} 2x dy = \frac{2}{3}$

$$E(Y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} y f(x, y) dx dy = \int_0^1 dx \int_x^{2x} 2y dy = 1$$

$$(2) E(XY) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy f(x, y) dx dy = \int_0^1 dx \int_x^{2x} 2xy dy = \frac{2}{4}$$

$$\text{cov}(X, Y) = E(XY) - EX \cdot EY = \frac{2}{4} - \frac{2}{3} \times 1 = -\frac{1}{6}$$

93. (1) $E(X) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x f(x, y) dx dy = \int_0^1 dx \int_{-x}^x x dy = \frac{2}{3}$

$$E(Y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} y f(x, y) dx dy = \int_0^1 dx \int_{-x}^x y dy = 0$$

$$(2) E(XY) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy f(x, y) dx dy = \int_0^1 dx \int_{-x}^x xy dy = 0$$

$$\therefore \text{cov}(X, Y) = E(XY) - EX \cdot EY = 0$$



$$E(X) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x f(x, y) dx dy = \frac{1}{8} \int_0^2 x dx \int_0^2 (x+y) dy = \frac{7}{6}$$

$$E(Y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} y f(x, y) dx dy = \frac{1}{8} \int_0^2 dx \int_0^2 y (x+y) dy = \frac{7}{6}$$

$$E(XY) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy f(x, y) dx dy = \frac{1}{8} \int_0^2 x dx \int_0^2 y (x+y) dy = \frac{4}{3}$$

$$\text{cov}(X, Y) = E(XY) - E(X) \cdot E(Y) = \frac{4}{3} - \frac{7}{6} \times \frac{7}{6} = -\frac{1}{36}$$

$$E(X^2) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x^2 f(x, y) dx dy = \frac{1}{8} \int_0^2 x^2 dx \int_0^2 (x+y) dy = \frac{5}{3}$$

$$E(Y^2) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} y^2 f(x, y) dx dy = \frac{1}{8} \int_0^2 dx \int_0^2 (x+y) y^2 dy = \frac{5}{3}$$

$$D(X) = E(X^2) - (E(X))^2 = \frac{5}{3} - \left(\frac{7}{6}\right)^2 = \frac{11}{36}$$

$$D(Y) = E(Y^2) - (E(Y))^2 = \frac{5}{3} - \left(\frac{7}{6}\right)^2 = \frac{11}{36}$$

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{D(X) \cdot D(Y)}} = -\frac{1}{11}$$

$$\begin{aligned} E(X^2 Y^2) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x^2 y^2 f(x, y) dx dy \\ &= \frac{1}{8} \int_0^2 dx \int_0^2 x^2 y^2 (x+y) dy \\ &= \frac{8}{3} \end{aligned}$$

$$\therefore D(XY) = E(X^2 Y^2) - [E(XY)]^2 = \frac{8}{3} - \left(\frac{4}{3} \times \frac{4}{3}\right) = \frac{8}{9}$$

95. (1) $f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \int_0^x z dy = zx, 0 < x < 1$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \int_y^1 z dx = z(1-y), 0 < y < 1$$

$$\therefore f_X(x) = \begin{cases} zx, & 0 < x < 1 \\ 0, & \text{其他} \end{cases}, f_Y(y) = \begin{cases} z(1-y), & 0 < y < 1 \\ 0, & \text{其他} \end{cases} \quad (3) \because f(x, y) \neq f_X(x) \cdot f_Y(y)$$

$$\begin{aligned} (2) f_{Y|X}(y|x) &= \frac{f(x, y)}{f_X(x)} = \begin{cases} \frac{1}{x}, & y \leq x, 0 < x < 1 \\ 0, & \text{其他} \end{cases} \\ f_{X|Y}(x|y) &= \frac{f(x, y)}{f_Y(y)} = \begin{cases} \frac{1}{1-y}, & y \leq x, 0 < x < 1 \\ 0, & \text{其他} \end{cases} \end{aligned}$$



96. (1) $f_X(x) = \int_{-\infty}^{+\infty} f(x,y) dy = \int_{x^2}^1 \frac{21}{4} x^2 y dy$
 $= \frac{21}{8} (1-x^4) x^2, -1 \leq x \leq 1$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x,y) dx = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{21}{4} x^2 y dx$$

$$= \frac{21}{6} y^{\frac{5}{2}}, 0 \leq y \leq 1$$

$$\therefore f_X(x) = \begin{cases} \frac{21}{8} x^2 (1-x^4), & -1 \leq x \leq 1 \\ 0, & \text{其他} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{21}{6} y^{\frac{5}{2}}, & 0 \leq y \leq 1 \\ 0, & \text{其他} \end{cases}$$

(2) $\because f(x,y) \neq f_X(x) \cdot f_Y(y), \therefore X$ 与 Y 不独立.

(3) $f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \begin{cases} \frac{1}{2} y^{-\frac{3}{2}}, & x^2 \leq y \leq 1 \\ 0, & \text{其他} \end{cases}$

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \begin{cases} \frac{1}{2} \frac{y}{(1-x^4)}, & x^2 < y \leq 1 \\ 0, & \text{其他} \end{cases}$$

97. (1) $E(X - \frac{1}{2}) = \int_{-\infty}^{+\infty} (x - \frac{1}{2}) f(x) dx = \int_0^1 (x^2 - \frac{1}{4}) dx = \frac{1}{12}$

(2) $P\{X > \frac{2}{3}\} = \int_{\frac{2}{3}}^1 f(x) dx = \int_{\frac{2}{3}}^1 (x + \frac{1}{2}) dx = \frac{4}{9}$

$$Y \sim B(n, p) = B(9, \frac{4}{9})$$

$$\therefore E(Y) = np = 9 \times \frac{4}{9} = 4, D(Y) = np(1-p) = \frac{20}{9}$$

$$E(Y^2) = D(Y) + [E(Y)]^2 = \frac{20}{9} + 16 = \frac{164}{9}$$



$$\begin{aligned} (1) \quad 1 &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x,y) dx dy = \int_0^1 dx \int_0^{1-x} 24xy dy \\ &= 24 \int_0^1 x dx \int_0^{1-x} y dy = \frac{24}{2} \int_0^1 x (1-x)^2 dx \Rightarrow 24 \end{aligned}$$

$$(2) \quad f_X(x) = \int_{-\infty}^{+\infty} f(x,y) dy = \int_0^{1-x} 24xy dy = 12x(1-x)^2, 0 \leq x \leq 1$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x,y) dx = \int_0^{1-y} 24xy dx = 12y(1-y)^2, 0 \leq y \leq 1$$

$$\therefore f_X(x) = \begin{cases} 12x(1-x)^2, & 0 \leq x \leq 1 \\ 0, & \text{其他} \end{cases} \quad f_Y(y) = \begin{cases} 12y(1-y)^2, & 0 \leq y \leq 1 \\ 0, & \text{其他} \end{cases}$$

$$(3) \quad f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \begin{cases} \frac{2y}{(1-x)^2}, & 0 \leq x \leq 1, 0 \leq y \leq 1-x \\ 0, & \text{其他} \end{cases}$$

$$(4) \quad \because f(x,y) \neq f_X(x) \cdot f_Y(y), \therefore X \text{ 与 } Y \text{ 不独立.}$$

$$99. (1) \quad f_X(x) = \int_{-\infty}^{+\infty} f(x,y) dy = \begin{cases} 4(1-x)^3, & 0 \leq x \leq 1 \\ 0, & \text{其他} \end{cases}$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x,y) dx = \begin{cases} 8y[1-y - \frac{1}{2}(1-y)^2], & 0 \leq y \leq 1 \\ 0, & \text{其他} \end{cases}$$

$$(2) \quad f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \begin{cases} \frac{2y}{(1-x)^2}, & 0 \leq x \leq 1, 0 \leq y \leq 1-x \\ 0, & \text{其他} \end{cases}$$

$$(3) \quad \because f(x,y) \neq f_X(x) \cdot f_Y(y), \therefore X \text{ 与 } Y \text{ 不独立.}$$



100. 设 X 为“正态分布”的函数, n 为总函数, 则 $X \sim B(n, 0.5)$

$$EX = np = 0.5n, \quad DX = np(1-p) = 0.25n, \quad \sqrt{DX} = 0.5\sqrt{n}$$

$$\begin{aligned} P\{24 \leq \frac{X}{n} \leq 26\} &= P\{0.48 \leq X \leq 0.6n\} \\ &= P\left\{\frac{0.48n - 0.5n}{0.5\sqrt{n}} \leq \frac{X - 0.5n}{0.5\sqrt{n}} \leq \frac{0.6n - 0.5n}{0.5\sqrt{n}}\right\} \\ &= P\{-0.2\sqrt{n} \leq \frac{X - 0.5n}{0.5\sqrt{n}} \leq 0.2\sqrt{n}\} = \Phi(0.2\sqrt{n}) - \Phi(-0.2\sqrt{n}) \\ &= 2\Phi(0.2\sqrt{n}) - 1 \geq 0.9, \text{ 即 } \Phi(0.2\sqrt{n}) \geq 0.95 = \Phi(1.65) \\ &\Rightarrow 0.2\sqrt{n} \geq 1.65 \Rightarrow n \geq 68.0625, \therefore n = 69. \end{aligned}$$

101. 设零件产品的重量为 $X_i, i=1, 2, \dots, 100$, $EX_i = 100, \sqrt{DX_i} = 10$.

$$\begin{aligned} \text{则 } P\left\{\sum_{i=1}^{100} X_i > 10200\right\} &= 1 - P\left\{\sum_{i=1}^{100} X_i \leq 10200\right\} \\ &= 1 - P\left\{\frac{\sum_{i=1}^{100} X_i - 10000}{100} \leq \frac{10200 - 10000}{100}\right\} = 1 - \Phi(2) = 0.02275. \end{aligned}$$

102. 设 X 为没有损坏的部件数, 则 $X \sim B(100, 0.9)$

$$EX = np = 90, \quad DX = np(1-p) = 9, \quad \sqrt{DX} = 3$$

$$\begin{aligned} P\{\text{系统工作}\} &= P\{X \geq 85\} = P\left\{\frac{X - 90}{3} \geq \frac{85 - 90}{3}\right\} = P\left\{\frac{X - 90}{3} \geq -\frac{5}{3}\right\} \\ &= 1 - P\left\{\frac{X - 90}{3} < -\frac{5}{3}\right\} = 1 - \Phi\left(-\frac{5}{3}\right) = \Phi\left(\frac{5}{3}\right) \approx \Phi(1.67) = 0.9525 \end{aligned}$$

103. 设 X 为该市一年内发生重大人身事故的人数, 则 $X \sim B(5000, 0.005)$

$$EX = np = 25, \quad DX = 25 \times 0.995 = 24.875.$$

$$\begin{aligned} P\{20 \leq 80 - 2X \leq 40\} &= P\{20 \leq X \leq 30\} = P\left\{\frac{20 - 25}{\sqrt{24.875}} \leq \frac{X - 25}{\sqrt{24.875}} \leq \frac{30 - 25}{\sqrt{24.875}}\right\} \\ &= 2\Phi\left(\frac{5}{\sqrt{24.875}}\right) - 1 \approx 2\Phi(1) - 1 = 0.6826. \end{aligned}$$



104. 设第 i 位顾客的消费额为 $X_i, i=1, 2, \dots, 100$, 且 X_i 服从 $[6, 15]$ 上的均匀分布. $EX_i = 7.8, DX_i = \frac{(15-6)^2}{12} = 1728$.

$$P\left\{\left|\sum_{i=1}^{100} X_i - 100 \times 7.8\right| \leq 500\right\} = P\left\{\left|\frac{\sum_{i=1}^{100} X_i - 780}{10\sqrt{1728}}\right| \leq \frac{500}{10\sqrt{1728}}\right\}$$

$$= 2\Phi(1.2) - 1 = 0.7698.$$

105. 设开灯的教室数为 X , 则 $X \sim B(100, 0.8)$.

$$EX = np = 80, DX = np(1-p) = 16.$$

$$P\{60 \leq X \leq 80\} = P\left\{\frac{60-80}{4} \leq \frac{X-80}{4} \leq \frac{80-80}{4}\right\}$$

$$= \Phi(0) - \Phi(-5) = \Phi(0) + \Phi(5) - 1 = 0.5$$

106. 考试成绩为 $X \sim N(\mu, \sigma^2)$. 因为 $P\{X \geq 60\} = 25\%$.

$$P\{X > 80\} = 3\%. \text{ 所以 } \begin{cases} P\{X < 60\} = 0.75 \\ P\{X \leq 80\} = 0.97 \end{cases}$$

$$\text{即 } \begin{cases} P\left\{\frac{X-\mu}{\sigma} < \frac{60-\mu}{\sigma}\right\} = 0.75 = \Phi(0.68) \\ P\left\{\frac{X-\mu}{\sigma} < \frac{80-\mu}{\sigma}\right\} = 0.97 = \Phi(1.89) \end{cases} \Rightarrow \begin{cases} \frac{60-\mu}{\sigma} = 0.68 \\ \frac{80-\mu}{\sigma} = 1.89 \end{cases}$$

$$\Rightarrow \begin{cases} \mu = 48.8 \\ \sigma = 16.5 \end{cases} \therefore P\{X > 70\} = 1 - P\{X \leq 70\} = 1 - P\left\{\frac{X-\mu}{\sigma} \leq \frac{70-\mu}{\sigma}\right\} \\ = 1 - \Phi\left(\frac{70-\mu}{\sigma}\right) \approx 1 - \Phi(1.28) = 0.1003$$



107. (1) 矩估计法:

$$EX = \theta^2 + 4\theta(1-\theta) + 3(1-\theta)^2 = 3 - 2\theta$$

$$\bar{x} = \frac{1+2+1}{3} = \frac{4}{3}$$

$$\text{令 } EX = \bar{x} \Rightarrow 3 - 2\theta = \frac{4}{3} \Rightarrow \hat{\theta} = \frac{5}{6}$$

故 θ 的矩估计值为 $\frac{5}{6}$.

(2) 最大似然估计法:

$$\text{似然函数 } L(\theta) = \theta^2 \cdot 2\theta(1-\theta) \cdot \theta^2 = 2\theta^5(1-\theta)$$

$$\text{对数似然函数 } \ln L(\theta) = \ln 2 + 5\ln \theta + \ln(1-\theta)$$

$$\text{求导并令其为零: } \frac{d \ln L(\theta)}{d\theta} = \frac{5}{\theta} - \frac{1}{1-\theta} = 0$$

$$\text{解得: } \hat{\theta}_1 = \frac{5}{6}. \quad \text{即最大似然估计值为 } \frac{5}{6}.$$

108. (1) 矩估计法:

$$EX = 2\theta(1-\theta) + 2\theta^2 + 3(1-2\theta) = 3 - 4\theta$$

$$\bar{x} = \frac{3+1+3+0+3+1+2}{7}$$

$$\text{令 } EX = \bar{x} \text{ 得: } 3 - 4\theta = \frac{13}{7}, \hat{\theta} = \frac{2}{7}.$$

即 θ 的矩估计值为 $\frac{2}{7}$.

(2) 最大似然估计法:

$$L(\theta) = (1-2\theta)^3 [2\theta(1-\theta)]^2 \cdot \theta^2 \cdot \theta^2 = 4\theta^6(1-\theta)^2(1-2\theta)^3$$

$$\ln L(\theta) = \ln 4 + 6\ln \theta + 2\ln(1-\theta) + 3\ln(1-2\theta)$$

$$\frac{d \ln L(\theta)}{d\theta} = \frac{6}{\theta} - \frac{2}{1-\theta} - \frac{6}{1-2\theta} = 0, \quad \hat{\theta}_1 = 0.068676$$
$$= \frac{13 - \sqrt{132}}{22}$$



109. (1) 矩估计法:

$$E(X) = \int_0^1 x f(x) dx = \int_0^1 (\alpha+1) x^{\alpha+1} dx = \frac{\alpha+1}{\alpha+2}$$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\text{令 } E(X) = \bar{x}, \text{ 解得: } \alpha = \frac{2\bar{x}-1}{1-\bar{x}}$$

$$\therefore \theta \text{ 的矩估计值为 } \frac{2\bar{x}-1}{1-\bar{x}}, \text{ 矩估计量为 } \frac{2\bar{X}-1}{1-\bar{X}}.$$

(2) 最大似然估计法:

$$L(\alpha) = \prod_{i=1}^n (\alpha+1) x_i^{\alpha} = (\alpha+1)^n \left(\prod_{i=1}^n x_i \right)^{\alpha}, 0 < x_i < 1$$

$$\ln L(\alpha) = n \ln(\alpha+1) + \alpha \sum_{i=1}^n \ln x_i$$

$$\frac{d \ln L(\alpha)}{d \alpha} = \frac{n}{\alpha+1} + \sum_{i=1}^n \ln x_i = 0, \hat{\alpha}_L = -\frac{n}{\sum_{i=1}^n \ln x_i} - 1$$

110. 令 $L(p) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i} = p^{\sum_{i=1}^n x_i} (1-p)^{n - \sum_{i=1}^n x_i}$

$$\ln L(p) = \sum_{i=1}^n x_i \ln p + (n - \sum_{i=1}^n x_i) \ln (1-p)$$

$$d \ln L(p) = \sum_{i=1}^n x_i \cdot \frac{1}{p} - \frac{1}{1-p} \cdot (n - \sum_{i=1}^n x_i) = 0$$

$$\text{解得 } \hat{p} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}.$$

111. (1) $L(\lambda) = \left(\frac{\lambda^4}{4!} e^{-\lambda} \right)^4 \left(\frac{\lambda^2}{2!} e^{-\lambda} \right)^2 \left(\frac{\lambda^0}{0!} e^{-\lambda} \right)^2 \left(\frac{\lambda^3}{3!} e^{-\lambda} \right) \cdot \frac{\lambda^5}{5!} e^{-\lambda}$

$$= \frac{1}{2880} \lambda^{16} e^{-10\lambda}$$

$$d \ln L(\lambda) = \ln \frac{1}{2880} + 16 \ln \lambda - 10\lambda$$

$$\frac{d \ln L(\lambda)}{d \lambda} = \frac{16}{\lambda} - 10 = 0 \Rightarrow \hat{\lambda} = 1.6$$

(2) $P\{X=4\} = \frac{1.6^4}{4!} e^{-1.6} = 0.055132$



$$12. L(\theta) = \prod_{i=1}^n \theta C^{\theta} x_i^{-(\theta+1)} = \theta C^n \left(\prod_{i=1}^n x_i \right)^{-(\theta+1)}, \quad x_i > C$$

$$\ln L(\theta) = n \ln \theta + n \ln C - (\theta+1) \sum_{i=1}^n \ln x_i$$

$$\frac{d \ln L(\theta)}{d\theta} = \frac{n}{\theta} + n \ln C - \sum_{i=1}^n \ln x_i = 0$$

$$\text{解得: } \hat{\theta} = \frac{n}{\sum_{i=1}^n \ln x_i - n \ln C}$$

$$113. L(\theta) = \prod_{i=1}^n \frac{x_i e^{-\frac{x_i}{2\theta}}}{\theta} = \frac{\prod_{i=1}^n x_i \cdot e^{-\frac{1}{2\theta} \sum_{i=1}^n x_i}}{\theta^n}$$

$$\ln L(\theta) = \sum_{i=1}^n \ln x_i - \frac{1}{2\theta} \sum_{i=1}^n x_i - n \ln \theta$$

$$\frac{d \ln L(\theta)}{d\theta} = -\frac{1}{2\theta^2} \sum_{i=1}^n x_i - \frac{n}{\theta} = 0, \quad \hat{\theta} = \frac{\sum_{i=1}^n x_i}{2n}$$

$$114. L(\theta) = \prod_{i=1}^n 2 e^{-2(x_i - \theta)} = 2^n e^{-2 \sum_{i=1}^n x_i + 2n\theta}, \quad x_i > 0$$

$$\ln L(\theta) = n \ln 2 - 2 \sum_{i=1}^n x_i + 2n\theta$$

$$\frac{d \ln L(\theta)}{d\theta} = 2n > 0, \text{ 所以 } \ln L(\theta) \text{ 随 } \theta \text{ 的增加而增加,}$$

$$\therefore \theta < x_i (i=1, 2, \dots, n), \quad \hat{\theta} = \min \{x_1, x_2, \dots, x_n\}$$

$$115. EX = \int_{-\infty}^{+\infty} x f(x; \theta) dx = \int_0^{\theta} 0 dx + \int_{\theta}^{+\infty} (x - \theta) dx = \frac{\theta}{2} - \theta$$

$$\text{令 } EX = \bar{X} \text{ 得 } \theta \text{ 的矩估计量为 } \hat{\theta} = \frac{\theta}{2} - \bar{X}$$

设 N 为样本值 $x_1, x_2, \dots, x_n$ 中小于 θ 的个数, 则

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta) = \theta^N (1-\theta)^{n-N}$$

$$\ln L(\theta) = N \ln \theta + (n-N) \ln (1-\theta), \quad \frac{d \ln L(\theta)}{d\theta} = \frac{N}{\theta} + \frac{n-N}{1-\theta} = 0$$

$$\hat{\theta} = \frac{N}{n}$$



$$16. P\{|\bar{X}-2|\leq 0.24\} = P\left\{\left|\frac{\bar{X}-2}{2/5}\right|\leq \frac{0.24}{2/5}\right\} = 2\Phi(0.6)-1 = 0.4514$$

$$117. P\left\{\frac{S^2}{\sigma^2}\leq 2.041\right\} = P\left\{\frac{S^2}{\sigma^2}(n-1)\leq 2.041\times(n-1)\right\} \\ = ~~0.01~~ P\left\{\frac{S^2(n-1)}{\sigma^2}\leq 30.615\right\} \approx P\left\{\frac{15S^2}{\sigma^2}\leq \chi_{0.01}^2(15)\right\} \\ = 1-0.01 = 0.99.$$

$$118. P\{\bar{X}\leq 10.48\} = P\left\{\frac{\bar{X}-\mu}{S/\sqrt{n}}\leq \frac{10.48-10}{0.9/\sqrt{6}}\right\} \\ = P\left\{\frac{\bar{X}-\mu}{S/\sqrt{n}}\leq 2.4\right\} \approx P\left\{\frac{\bar{X}-\mu}{S/\sqrt{n}}\leq t_{0.025}(15)\right\} = 1-0.025 = 0.975$$

119. $n=9, \bar{X}=9.62, S^2=0.36, 1-\alpha=0.95, \alpha/2=0.025$
 选取 $\frac{\bar{X}-\mu}{S/\sqrt{n}}$ 为枢轴变量, 且 $\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n-1)$

由 $P\{-t_{\alpha/2}(n-1) < \frac{\bar{X}-\mu}{S/\sqrt{n}} < t_{\alpha/2}(n-1)\} = 1-\alpha$ 得:

置信区间为: $(\bar{X} - t_{\alpha/2}(n-1) \cdot \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2}(n-1) \cdot \frac{S}{\sqrt{n}})$

$$= (9.62 - 2.306 \times \frac{0.6}{3}, 9.62 + 2.306 \times \frac{0.6}{3})$$

$$= (9.1596, 10.0604)$$

120. $\bar{X}=150, n=100, \sigma=22, 1-\alpha=0.95, \alpha/2=0.025$.

选 $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ 为枢轴变量, 由 $P\{-U_{\alpha/2} < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < U_{\alpha/2}\} = 1-\alpha$ 得

置信区间 $(\bar{X} - U_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + U_{\alpha/2} \frac{\sigma}{\sqrt{n}}) = (145.688, 154.312)$



121. $n=25$, $\bar{X}=186$, $S=12$, $1-\alpha=0.9$, $\alpha/2=0.05$, μ 未知

选取 $\frac{(n-1)S^2}{\sigma^2}$ 为枢轴变量, 且 $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)}$

由 $P\left\{\chi^2_{1-\alpha/2} < \frac{(n-1)S^2}{\sigma^2} < \chi^2_{\alpha/2}\right\} = 1-\alpha$ 得

σ^2 的置信区间为 $\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2}(n-1)}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}(n-1)}\right) =$

$$\left(\frac{24 \times 12^2}{36.42}, \frac{24 \times 12^2}{13.85}\right) \approx (94.89291, 249.53069)$$

122. (1) $\bar{X} = \frac{1}{4} \sum_{i=1}^4 X_i$, $E\bar{X} = \frac{1}{4} \sum_{i=1}^4 EX_i = 0$, $D\bar{X} = \frac{1}{4}$

$$(2) \frac{(X_1 + X_2)^2}{(X_3 - X_4)^2} = \frac{\left(\frac{X_1 + X_2}{\sqrt{2}}\right)^2}{\left(\frac{X_3 - X_4}{\sqrt{2}}\right)^2} \sim F(1, 1)$$

123. (1) $E\bar{X} = 0$, $D\bar{X} = \frac{1}{n}$

$$(2) \frac{4X_1}{\sqrt{X_2^2 + X_3^2 + \dots + X_{17}^2}} = \frac{X_1}{\sqrt{\frac{X_2^2 + X_3^2 + \dots + X_{17}^2}{16}}} \sim t(16)$$

124. (1) $E(T_1) = \left(\frac{2}{6} + \frac{2}{3}\right)\mu = \mu$, $E(T_2) = 2\mu$, $E(T_3) = \mu$.

故下, T_3 是 μ 的无偏估计量.

$$(2) D(T_1) = \left(\frac{1}{36} + \frac{1}{36} + \frac{1}{9} + \frac{1}{9}\right)\sigma^2 = \frac{10}{36}\sigma^2$$

$$D(T_3) = \frac{4}{16}\sigma^2 < D(T_1), \therefore T_3 \text{ 更有效.}$$



5. (1)

$X \backslash Y$	-1	0	1	$P_{i\cdot}$
0	0	$\frac{1}{3}$	0	$\frac{1}{3}$
1	$\frac{1}{3}$	0	$\frac{1}{3}$	$\frac{2}{3}$
$P_{\cdot j}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	1

(2) $P\{X=0, Y=-1\} = 0 \neq P\{X=0\} \cdot P\{Y=-1\}$

$\therefore X$ 与 Y 不独立.

126. $P\{X=0\} = \frac{C_8^1}{C_{10}^1} = \frac{4}{5} = \frac{36}{45}$

$P\{X=2\} = \frac{C_2^1 C_8^1}{C_{10}^1 C_9^1} = \frac{8}{45}$

$P\{X=3\} = \frac{C_2^1 C_1^1 C_8^1}{C_{10}^1 C_9^1 C_8^1} = \frac{1}{45}$

$EX = 1 \times \frac{36}{45} + 2 \times \frac{8}{45} + 3 \times \frac{1}{45} = \frac{55}{45} = \frac{11}{9}$

~~EX =~~

$EX^2 = 1^2 \times \frac{36}{45} + 2^2 \times \frac{8}{45} + 3^2 \times \frac{1}{45} = \frac{77}{45}$

$DX = EX^2 - (EX)^2 = \frac{77}{45} - \frac{55^2}{45^2} = \frac{22}{45}$

127. $P\{X=0\} = 0.99 \times 0.98 \times 0.97 = 0.941084$

$P\{X=1\} = 0.057818$

$P\{X=2\} = 0.001082$

$P\{X=3\} = 0.01 \times 0.02 \times 0.03 = 0.000006$

$EX = 0.057818 + 2 \times 0.001082 + 3 \times 0.000006 = 0.06$

$EX^2 = 0.057818 + 4 \times 0.001082 + 9 \times 0.000006 = 0.0622$

$DX = EX^2 - (EX)^2 = 0.0586$



$$128. (1) 1 = \int_0^{\frac{1}{2}} (ax^2 + x) dx = \left(\frac{a}{3} x^3 + \frac{1}{2} x^2 \right) \Big|_0^{\frac{1}{2}} = \frac{a}{24} + \frac{1}{8} \Rightarrow a = 2$$

$$(2) F(x) = \begin{cases} 0, & x < 0 \\ \frac{2}{3} x^3 + \frac{1}{2} x^2, & 0 \leq x \leq \frac{1}{2} \\ 1, & x > \frac{1}{2} \end{cases} = \begin{cases} 0, & x < 0 \\ \frac{2}{3} x^3 + \frac{1}{2} x^2, & 0 \leq x \leq \frac{1}{2} \\ 1, & x > \frac{1}{2} \end{cases}$$

$$(3) P\left(\frac{1}{7} < X < 2\right) = F(2) - F\left(\frac{1}{7}\right) = 1 - \frac{3}{49} = \frac{46}{49}$$

$$129. (1) 1 = \int_1^2 a \left(1 - \frac{1}{x^2}\right) dx = a \left(1 - \frac{1}{2}\right) = \frac{a}{2} \Rightarrow a = 2$$

$$(2) F(x) = \begin{cases} 0, & x < 1 \\ 2\left(x - \frac{1}{x}\right) - 4, & 1 \leq x \leq 2 \\ 1, & x > 2. \end{cases}$$

$$(3) P\left(\frac{3}{2} \leq X \leq 2\right) = F(2) - F\left(\frac{3}{2}\right) = 1 - \frac{1}{3} = \frac{2}{3}$$

